Central Limit theorem

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The Central Limit Theorem (CLT) is a fundamental concept in probability theory and statistics. It states that, under certain conditions, the distribution of the sum (or average) of a large number of independent, identically distributed random variables will be approximately normally distributed, regardless of the original distribution of the variables themselves.

Here are the key points and implications of the Central Limit Theorem:

- **Independent and Identically Distributed (I.I.D.) Random Variables:** The CLT applies to a set of random variables that are independent and identically distributed. This means that each random variable is drawn from the same distribution and is not influenced by any other random variable in the set.
- **Sample Size:** The CLT holds particularly well as the sample size increases. The larger the sample size, the closer the distribution of the sample mean (or sum) will be to a normal distribution.
- **Approximation:** The CLT allows us to approximate the distribution of the sample mean (or sum) even when the original distribution of the individual random variables is not normal. This is extremely powerful because the normal distribution is well understood and has many convenient properties for statistical inference.
- **Normal Distribution:** According to the CLT, as the sample size tends to infinity, the distribution of the sample mean (or sum) becomes exactly normal, regardless of the original distribution of the individual random variables. For practical purposes, even relatively small sample sizes often result in a good approximation to the normal distribution.

Central Limit theorem

Central Limit Theorem (Linderberg-Levy's Form)

If $X_1, X_2, \cdots, X_n, \cdots$ be a sequence of independent identically distributed RVs with $E(X_i) = \mu$ and $Var(X_i) = \sigma^2$, $i = 1, 2, \cdots$ and if $S_n = X_1 + X_2 + \cdots + X_n$, then under certain general conditions, S_n follows a normal distribution with mean $n\mu$ and variance $n\sigma^2$ as n tends to infinity.

Property 1:

If $\bar{X} = \frac{1}{n}(X_1 + X_2 + \cdots + X_n)$, then $E(\bar{X}) = \mu$ and $Var(\bar{X}) = \dfrac{1}{n^2}(n\sigma^2) = \dfrac{\sigma^2}{n}$

$\therefore \quad \bar{X}$ follows $N\left(\mu, \frac{\sigma}{\sqrt{n}}\right)$ as $n \to \infty$

Worked Problem 1

The lifetime of a certain brand of an electric bulb may be considered a RV with mean 1200 h and Standard deviation 250 h. Find the probability, using central limit theorem, that the average lifetime of 60 bulbs exceeds 1250 h.

Solution:
Let X_i represent the lifetime of the bulb.

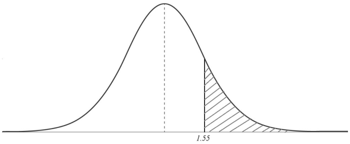
$$E(X_i) = 1200 \quad \text{and} \quad Var(X_i) = 250^2$$

Let $\bar{X}$ denote the mean lifetime of 60 bulbs.
By Property 1 of Lindeberg-Levy form of CLT, $\bar{X}$ follows $N\left(1200, \frac{250}{\sqrt{60}}\right)$

$$P(\bar{X} > 1250) = P\left(\frac{\bar{X} - 1200}{\frac{250}{\sqrt{60}}} > \frac{1250 - 1200}{\frac{250}{\sqrt{60}}}\right)$$
$$= P\left(z > \frac{\sqrt{60}}{5}\right)$$
$$= P(z > 1.55),$$

where z is the standard normal variable

$$= 0.0606 \text{ (from the table of areas under normal curve)}$$



Worked Problem 2

A distribution with unknown μ has variance equal to 1.5. Use central limit theorem to find how large a sample should be taken from the distribution in order that the probability will be at least 0.95 that the sample mean will be within 0.5 of the population mean.

Solution: Let n be the size of the sample, a typical number of which is X_i
Given: $E(X_i) = \mu$ and $Var(X_i) = 1.5$.
Let $\bar{X}$ denote the sample mean. By Corollary under CLT,

$$\bar{X} \text{ follows } N\left(\mu, \frac{\sqrt{1.5}}{\sqrt{n}}\right)$$

We have to find n such that

$$\begin{aligned}
& P\{\mu - 0.5 < \bar{X} < \mu + 0.5\} \geq 0.95 \\
\text{i.e.,}\quad & P\{-0.5 < \bar{X} - \mu < 0.5\} \geq 0.95 \\
\text{i.e.,}\quad & P\{|\bar{X} - \mu| < 0.5\} \geq 0.95 \\
\text{i.e.,}\quad & P\left\{\frac{|\bar{X} - \mu|}{\sqrt{\frac{1.5}{n}}} < \frac{0.5}{\sqrt{\frac{1.5}{n}}}\right\} \geq 0.95 \\
\text{i.e.,}\quad & P\{|z| < 0.4082\sqrt{n}\} \geq 0.95
\end{aligned}$$

where z is the standard normal variable.

Worked Problem 2 Cntd.,

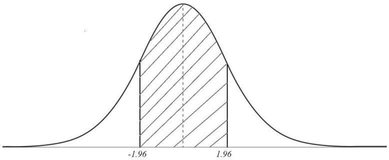
The least value of n is obtained from

$$P\{|z| < 0.4082\sqrt{n}\} = 0.95$$

From the table of areas under normal curve

$$P\{|z| < 1.96\} = 0.95$$

Therefore, least n is given by $0.4082\sqrt{n} = 1.96$, i.e., least $n = 24$.
Therefore the size of the sample must be at least 24.



Worked Problem 3

If $X_1, X_2, \cdots, X_n$ are Poisson variates with parameter $\lambda = 2$, use the central limit theorem to estimate $P(120 \leq S_n \leq 160)$, where $S_n = X_1 + X_2 + \cdots + X_n$ and $n = 75$.

Solution: $E(X_i) = \lambda = 2 \text{and} Var(X_i) = \lambda = 1.5$

By CLT, S_n follows $N(n\mu, \sigma\sqrt{n})$

ie., S_n follows $N(150, \sqrt{150})$

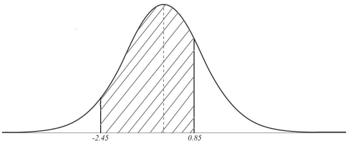
$$P\{120_n \leq 160\} = P\left\{\frac{-30}{\sqrt{150}} \leq \frac{S_n - 150}{\sqrt{150}} \leq \frac{10}{\sqrt{150}}\right\}$$

$= P\{-2.45 \leq z \leq 0.85\}$

where z is the standard normal variable.

$= 0.4927 + 0.2939$ (from the normal tables)

$= 0.7866$



Worked Problem 4

Suppose you roll a fair six-sided die 100 times and calculate the mean of the outcomes. What is the approximate distribution of the sample mean according to the central limit theorem?

Solution:
Each roll a fair six-sided die follows a discrete distribution with

$$\text{mean of } \mu = \frac{1+6}{2} = 3.5$$

$$\text{and a standard deviation} \sigma = \sqrt{\frac{(6-1+1)^2-1}{12}} = \sqrt{\frac{35}{12}} \approx 1.71$$

We are rolling the die 100 times, so according to the CLT, the sample mean will be approximately normally distributed with mean $\mu_x = \mu = 3.5$ and standard deviation $\sigma_x = \frac{\sigma}{\sqrt{n}} = \frac{1.71}{\sqrt{100}} = 0.171$
Therefore, the approximate distribution of the sample mean is

$$\bar{X} \sim \mathit{Normal}(3.5, 0.171^2)$$

Worked Problem 5

Suppose you have a population with mean (μ) of 50 and standard deviation (σ) of 10. You take a random sample of size 100 from this population. What are the mean and S.D. of the sampling distribution of the sample mean?

Solution:
According to CLT, the mean of the sampling distribution of the sample mean will be equal to the population mean which is 50 in this case.
The S.D. of the sampling distribution of the sample mean is given by

$$\sigma = \frac{\sigma}{\sqrt{n}}$$

where σ - population S.D and n - sample size
Substituting the given values,

$$\sigma = \frac{10}{\sqrt{100}} = 1$$

So the mean of the sampling distribution of the sample mean is 50 and S.D. is 1.

Worked Problem 6

Let $X_1, X_2, \cdots, X_{25}$ be i.i.d with the following PMF

$$P_X(k) = \begin{cases} 0.6 & \text{if } k = 1 \\ 0.4 & \text{if } k = -1 \\ 0 & \text{if } \textit{otherwise} \end{cases}$$

And let $Y = X_1 + X_2 + \cdots + X_n$, Using the CLT and continuity correction, estimates $P(4 \leq Y \leq 6)$

Solution:
We have

$$\begin{aligned} EX_i &= (0.6)(1) + (0.4)(-1) \\ &= \frac{1}{5} \\ EX_i^2 &= (0.6) + (0.4) \\ &= 1 \end{aligned}$$

Therefore,

$$Var(X_i) = 1 - \frac{1}{25} = \frac{24}{25}$$
$$\text{thus} \quad \sigma_{X_i} = \frac{2\sqrt{6}}{5}$$

Therefore,

$$EY = 25 \times \frac{1}{5} = 5$$
$$Var(Y) = 25 \times \frac{24}{25} = 24$$
$$\text{thus} \quad \sigma_Y = 2\sqrt{6}$$

$$\begin{aligned} P(4 \leq Y \leq 6) &= P(3.5 \leq Y \leq 6.5) \quad \text{(continuity correction)} \\ &= P\left(\frac{3.5-5}{2\sqrt{6}} \leq Y \leq \frac{6.5-5}{2\sqrt{6}}\right) \\ &= P\left(-0.3062 \leq \frac{Y-5}{2\sqrt{6}} \leq 0.3062\right) \\ &\approx \phi(0.3062) - \phi(-0.3062) \quad \text{by the CLT} \\ &= 2\phi(0.3062) - 1 \\ &\approx 0.2405 \end{aligned}$$

Worked Problem 7

The guaranteed average life of a certain type of electric light bulbs is 1000 hours with a standard deviation of 125 hours. It is proposed to sample the output so as to assure that 90% of the bulbs do not fall short of the guaranteed average by more than 2.5 %. What should be the minimum size of the sample? (The area under standard normal curve from $z = 0$ to $z = 1.28$ is 0.4000)

Solution:
Given: $\mu = 1000, \sigma = 125$. Let n be the size of the sample.
Now, "Short of the guaranteed average by more than 2.5%" $= (\bar{x} > 975)$.
By hypothesis, the probability of this event $= 90\% = \frac{9}{10}$
So,

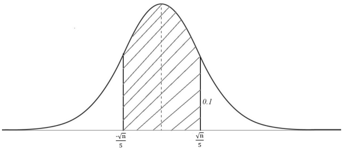
$$P(\bar{x} > 975) = \frac{9}{10} \tag{1}$$

The sampling distribution of $\bar{x}$ is normal distribution with mean $= \mu = 1000$ and S.d. $= \frac{\sigma}{\sqrt{n}} = \frac{125}{\sqrt{n}}$

So, $z = \dfrac{\bar{x} - 1000}{\frac{125}{\sqrt{n}}} = \dfrac{\sqrt{n}(\bar{x} - 1000)}{125}$ is a standard normal variate.

When $\bar{x} = 975,\ z = \dfrac{\sqrt{n}(975 - 1000)}{125} = -\dfrac{\sqrt{n}}{5}$

So, $P(\bar{x} > 975) = P\left(z > -\frac{\sqrt{n}}{5}\right)$



From (1), $P\left(z > -\frac{\sqrt{n}}{5}\right) = \frac{9}{10} = 0.9$ or $1 - P\left(z < -\frac{\sqrt{n}}{5}\right) = 0.9$

or $P\left(z < -\frac{\sqrt{n}}{5}\right) = 1 - 0.9 = 0.1$

By symmetry of normal curve,

$P\left(z > \frac{\sqrt{n}}{5}\right) = 0.1$ or $P\left(0 < z < \frac{\sqrt{n}}{5}\right) = 0.5 - 0.1 = 0.4$

So the area under standard normal curve enclosed between $z = 0$ and $z = \frac{\sqrt{n}}{5}$ is 0.4.

Worked Problem 8

Assume that 3000 male students of a university are normally distributed with mean 68.0 inches and standard deviation 3.0 inches. 80 samples consisting of 25 students each are obtained. In how many samples would you expect to find the mean between 66.8 and 68.3 inches.
Given area under standard normal curve enclosed between $z = -2$ and $z = 0$ is 0.4772 and between $z = 0$ and $z = 5$ is 0.1915

Solution:
The sample mean $\bar{X}$ is normally distributed with mean = population mean = 68 and

S.d. $= \dfrac{\sigma}{\sqrt{n}} = \dfrac{3}{\sqrt{25}} = 0.6$

So, $z = \dfrac{\bar{X} - 68}{0.6}$ is standard normal variate.

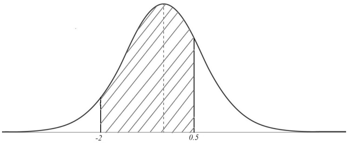
When $\bar{X} = 66.8$, $z = \dfrac{66.8 - 68}{0.6} = -2$

When $\bar{X} = 68.3$, $z = \dfrac{68.3 - 68}{0.6} = 0.5$. Therefore, Probability of $\bar{X}$ lying between 66.8 and 68.3

$$\begin{aligned}
&= P(66.8 < \bar{X} < 68.3) \\
&= P(-2 < z < 0.5) \\
&= P(-2 < z < 0) + P(0 < z < 0.5) \\
&= 0.4772 + 0.1915 = 0.6687
\end{aligned}$$

So, the expected number of required type of samples

$= 80 \times 0.6687 = 53.496 \approx 53$



Worked Problem 9

Distribution of marks scored in an examination is normal. Samples of four students are drawn and it is seen that the probability of the sample mean to be less than 61 is 0.44, to be more than 80 is 0.04. Find the mean and S.D of the distribution.
Given $\int_0^z \phi(t)dt = 0.06, 0.10, 0.46$ according as $z = 0.15, 0.25$ and 1.75

Solution:
Let mean $= \mu$ and S.D. $= \sigma$ of the population.
Since the population is normal so the sampling distribution of $\bar{x}$ (sample mean) is normal with mean $=$ population mean $= \mu$ and S.D. $= \frac{\sigma}{\sqrt{n}} = \frac{\sigma}{\sqrt{4}} = \frac{\sigma}{2}$
By Problem

$$P(\bar{x} < 61) = 0.44 \tag{2}$$
$$P(\bar{x} > 80) = 0.04 \tag{3}$$

Now, $z = \dfrac{\bar{x} - \mu}{\frac{\sigma}{2}} = \frac{2(\bar{x}-\mu)}{\sigma}$ has standard normal distribution.
When $\bar{x} = 61, z = \dfrac{2(61 - \mu)}{\sigma}$
When $\bar{x} = 80, z = \dfrac{2(80 - \mu)}{\sigma}$

$\therefore$ From (2), we have ,

$$P\left(z < \frac{2(61-\mu)}{\sigma}\right) = 0.44$$
$$P\left(\frac{2(61-\mu)}{\sigma} < z < 0\right) = 0.50 - 0.44 = 0.6$$
$$P\left(0 < z < -\frac{2(61-\mu)}{\sigma}\right) = 0.50 - 0.44 = 0.6$$
$$P\left(0 < z < -\frac{2(\mu-61)}{\sigma}\right) = 0.50 - 0.44 = 0.6 \tag{4}$$

We are given a data

$$P(0 < z < 0.15) = 0.6 \tag{5}$$

From (4) and (5) we have

$$\frac{2(\mu-61)}{\sigma} = .15$$
$$\text{or} \quad 2(\mu-61) = 0.15\sigma \tag{6}$$

Similarly from (3) and from other hypothesis we have

$$2(80-\mu) = 1.75\sigma \tag{7}$$

Solving (6) and (7) we get $\mu = 62.5$ and $\sigma = 20$